

B28: SQL SuperHero Program : Session 21

1 message

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See this query :

```
select H1.employee_id, H1.salary, H1.department_id,  
(case when H1.department_id = 90 then (select sum(H2.salary) from hr.employees H2 Where H2.Department_id = H1.Department_Id)  
  When H1.department_id = 60 then (select avg(H2.salary) from hr.employees H2 Where H2.Department_id = H1.Department_Id)  
  When H1.department_id = 100 then (select max(H2.salary) from hr.employees H2 Where H2.Department_id = H1.Department_Id)  
  Else 0  
End) Conditional_salary  
from hr.employees H1
```

If You see the above query, I am populating a new column as a conditional salary. My logic is if the department is 90 then give me the total salary for that department, if the department is 60 then give me avg salary of that department and if department is 100 then give me max salary of that department. You can see that case is so powerful, you can write the whole query here to populate the column.

```
1 select H1.employee_id, H1.salary, H1.department_id,  
2 (case when H1.department_id = 90 then (select sum(H2.salary) from hr.employees H2 Where H2.Department_id = H1.Department_Id)  
3   When H1.department_id = 60 then (select avg(H2.salary) from hr.employees H2 Where H2.Department_id = H1.Department_Id)  
4   When H1.department_id = 100 then (select max(H2.salary) from hr.employees H2 Where H2.Department_id = H1.Department_Id)  
5   Else 0  
6   End) Conditional_salary  
7 from hr.employees H1  
8  
9
```

EMPLOYEE_ID	SALARY	DEPARTMENT_ID	CONDITIONAL_SALARY
100	24000	90	58000
101	17000	90	58000
102	17000	90	58000
103	9000	60	5760
104	6000	60	5760
105	4800	60	5760
106	4800	60	5760
107	4200	60	5760
108	12008	100	12008
109	9000	100	12008

Case is so Powerful that every when and Then , you can write the full query if needed.

```
select H1.employee_id, h1.salary, h1.department_id,
(case when h1.department_id = 90 then (select count(*) from hr.countries where region_id = 1)
     when h1.department_id = 60 then (select avg(salary) from hr.employees H2 where h2.department_id = 90)
     when h1.department_id = 100 then (select max(salary) from hr.employees H2 where h2.department_id = 50)
     else 0
end) conditional_dalary
from hr.employees H1
```

case always support 1 thing as output.
only 1, not multiple values

Decode and case will be used in Order By as well.

3	SELECT
4	first_name,
5	last_name,
6	job_title
7	FROM
8	employees
9	ORDER BY
10	DECODE('F', 'F', first_name, 'L', last_name, 'J', job_title);

FIRST_NAME	LAST_NAME	JOB_TITLE
Aaron	Patterson	Stock Clerk
Abigail	Palmer	Shipping Clerk
Albert	Watson	Accountant
Alex	Sanders	Purchasing Clerk

If you see the above example, data is sorted by First_name. Why? In Order By clause we written that **Decode('F','F',first_name,'L',last_name,'J',job_title)**

It means if 'F' is 'F' then First_name, If 'F' is 'L' then Last_name, If 'F' is 'J' then Job_title. so first condition is satisfy and it order by first_name

3	SELECT
4	first_name,
5	last_name,
6	job_title
7	FROM
8	employees
9	ORDER BY
10	DECODE('J', 'F', first_name, 'L', last_name, 'J', job_title);

FIRST_NAME	LAST_NAME	JOB_TITLE
Tyler	Ramirez	Accountant
Albert	Watson	Accountant
Elli	James	Accountant
Elli	Brooks	Accountant
Ryan	Gray	Accountant

ordered

=====

```
create table country (name varchar2(20));
```

```
Insert Into country values ('England');
```

Insert Into country values ('Nigeria');
Insert Into country values ('Pakistan');
Insert Into country values ('India');
Insert Into country values ('Australia');
Insert Into country values ('Berlin');

select * from Country Order By Name;

NAME
Australia
Berlin
England
India
Nigeria
Pakistan

Download CSV
6 rows selected.

Question: I want output as India will come first and then all will come in Order.

There is a condition that If Name is India then it should be first else all other country name in order.

Select * From Country Order by case when name = 'India' then 'AAAA' else name end;

```
select * from country order by case when name = 'India' then 'AAAA' else name end
```

NAME
India
Australia
Berlin
England
Nigeria
Pakistan

6 rows selected.

case in order by

What exactly we did in the above query? We want India, should come before Australia, menas anyway, we just want a word before "Australia", lets take "AAAA", if we took "Ab" that also do the same. See the concept of Order By. We just need something which can be treated before

"Australia" in order by determination

```
select * from country order by case when name = 'India' then 'AA' else name end
```

NAME
India
Australia
Berlin
England
Nigeria
Pakistan

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6 rows selected.

AA

comes before
Australia

```
select * from country order by case when name = 'India' then 'Ar' else name end
```

NAME
India
Australia
Berlin
England
Nigeria
Pakistan

[Download CSV](#)

6 rows selected.

Ar is before Au

Suppose want India on 2nd

```
select * from country order by case when name = 'India' then 'Av' else name end
```

NAME
Australia
India
Berlin
England
Nigeria
Pakistan

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6 rows selected.

Av is come after Au

```
select * from country order by case when name = 'India' then 'Ba' else name end
```

NAME
Australia
India
Berlin
England
Nigeria
Pakistan

6 rows selected.

Ba before Berlin

```
select * from country order by case when name = 'India' then 'C' else name end
```

NAME
Australia
Berlin
India
England
Nigeria
Pakistan

— 3rd position, anything after
B and before E

6 rows selected.

```
select * from country order by case when name = 'India' then 'aa' else name end
```

NAME
Australia
Berlin
England
Nigeria
Pakistan
India

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6 rows selected.

ASCII value of
aa is
more

last

```
select * from country order by case when name = 'India' then 'AA' else name end
```

NAME
India
Australia
Berlin
England
Nigeria

FIRST

```
select ASCII('A') from dual
```

ASCII('A')
65

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```
select ASCII('a') from dual
```

a > A

ASCII('A')
97

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Suppose there is a twist and I want India on 1st position and England on 2nd position then what to do?

Now, If you see the input Data and understand the requirement, means anyhow we have to fix as India should come on 1st and England on 2nd.

```
select * from country
```

```
order by case when name = 'India' then 'AAAA'
```

```
when name = 'England' Then 'Ar'
```

```
else name end
```



```
select * from country
order by case when name = 'India' then 'AAAA'
when name = 'England' Then 'Ar'
else name end
```

NAME
India
England
Australia
Berlin
Nigeria
Pakistan

6 rows selected.

india change
AAAA
↓
so go to #1

Fix.
Something
to come before
Australia

Practice these questions:

Write a query to classify customers into tiers based on their total purchase amount:

50,000 → 'Platinum'

25,000-50,000 → 'Gold'

10,000-25,000 → 'Silver'

< 10,000 → 'Bronze'

Table: Customers (CustomerID, CustomerName, TotalPurchase)

Write a query to show students' grades based on marks:

= 90 → 'A'

80-89 → 'B'

70-79 → 'C'

Below 70 → 'Fail'

Table: Students (StudentID, StudentName, Marks)

Write a query to categorize products based on price: 'Expensive' (>1000), 'Affordable' (500-1000), and 'Cheap' (<500).

Table: Products (ProductID, ProductName, Price)

Write a query to classify orders as 'High Priority' if order value is above 5000, 'Medium Priority' for 2000-5000, and

'Low Priority' for below 2000.

Table: Orders (OrderID, CustomerID, OrderAmount)

Write a query to display a discount column based on total sales. If sales are greater than 10,000, give 20% discount, otherwise 10%.

Table: Sales (SaleID, CustomerID, TotalAmount)

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